

Causal Disentanglement for Semantic-Aware Intent Learning in Recommendation

Xiangmeng Wang[✉], Qian Li[✉], Dianer Yu[✉], Peng Cui[✉], *Senior Member, IEEE*,
Zhichao Wang[✉], and Guandong Xu[✉], *Member, IEEE*

Abstract—Traditional recommendation models trained on observational interaction data have generated large impacts in a wide range of applications, it faces bias problems that cover users' true intent and thus deteriorate the recommendation effectiveness. Existing methods track this problem as eliminating bias for the robust recommendation, e.g., by re-weighting training samples or learning disentangled representations. The disentangled representation methods as the state-of-the-art eliminate bias by revealing cause-effect of the bias generation. However, how to design the semantic-aware and unbiased representations for users' true intents is largely unexplored. To bridge the gap, we are the first to propose an unbiased and semantic-aware disentanglement learning called *CaDSI* (**C**ausal **D**isentanglement for **S**emantics-Aware **I**ntent Learning) from a causal perspective. Particularly, CaDSI explicitly models the causal relations underlying recommendation task, and thus produces semantic-aware representations via disentangling users' true intents aware of specific item context. Moreover, the causal intervention mechanism is designed to eliminate confounding bias stemming from context information, which further aligns the semantic-aware representation with users' true intent. Extensive experiments and case studies both validate the robustness and interpretability of our proposed model.

Index Terms—Causal disentanglement learning, causal intervention, semantic-aware representation

1 INTRODUCTION

RECOMMENDER system (RS) has become a panacea for any scenario requiring personalized recommendations, to help users discover users' interested products from overwhelming alternatives. Early works mainly adopt collaborative filtering (CF) methods [1], [2] to model user preference on items, based on historical user-item interactions (e.g., ratings, clicks). However, such user-item interaction usually exhibits bias that is entangled with users' real interests, degrading the recommendation performance ultimately. For instance, in movie recommendations, users are more likely to watch movies that are watched by many people, which is however due to users' conformity to other people, rather than stemming from users' real interests [3], [4].

Therefore, it is essential to capture users' pure interests independent of the bias to build high-quality recommender models.

Most existing works on bias-aware recommendation can be placed into two categories. The first category adopts re-weighting strategies on the observed interaction samples, to imitate the scenario that samples are evenly distributed without bias [5], [6], [7], [8]. One major limitation of these methods is that they merely mitigate bias at the data level, but fail to answer the fundamental question: what are the root causes for bias amplification. Another category aims to disentangle users' true interest via inspecting cause-effect of the bias generation for the robust recommendation, which has recently gained much attention [4], [9], [10]. Particularly, these works usually design a specific causal graph attributing the bias to a confounder. For example, social network is in fact a confounder for exposure bias, since it influences both users' choice of movie watching and their ratings [9]. Apparently, the confounder introduces pseudo-intents, the ignorance of which misguides the learning of user's true intent. A widely used solution of these works is to learn the user representation that is forced to be independent of the confounder, to uncover the user's true intents for final downstream recommendations. Particularly, these works design a regularizer for the independence constraint via statistical measures like L_1 -inv, L_2 -inv [11] and distance correlation [8]. By explicitly disentangling the cause of a confounder, the representation uncovering users' true intent can be learned for final recommendations.

Although promising improvements have been observed, existing approaches on disentangling user intent from bias still suffer two limitations: First, most of them merely focus on user-item interactions, which however suffer sparse

- Xiangmeng Wang, Dianer Yu, and Guandong Xu are with Data Science and Machine Intelligence Lab, Faculty of Engineering and Information Technology, University of Technology Sydney, Ultimo, NSW 2007, Australia. E-mail: {xiangmeng.wang, dianer.yu-1}@student.uts.edu.au, gdxu.au@gmail.com.
- Qian Li is with the School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University, Bentley, WA 6102, Australia. E-mail: qli@curtin.edu.au.
- Peng Cui is with the Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China. E-mail: cuip@tsinghua.edu.cn.
- Zhichao Wang is with the School of Electrical Engineering and Telecommunications, University of New South Wales, Sydney, NSW 2052, Australia. E-mail: zchaoking@gmail.com.

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Fig. 1. An example of bias: movie *Harry Potter* (HP) contains only one aspect *Director*, however, *Actor* and *Type* are missing. The high rating of the user on movie *Harry Potter* trains a prediction model towards the user's preference on the director *Steve Kloves*. In contrast, we observed that movie *Racing with the Moon* (RWM) with the same director received very low ratings from the same user.

issues [12], [13], [14] that lead to the difficulty of learning effective user/item representations. Moreover, existing disentangled learning methods merely treat one user-item interaction record as an independent instance and neglect its rich context information. We claim that the rich context information in the form of heterogeneous information can help to disentangle and interpret semantic-aware intents of users for robust recommendations. For instance, higher-order path structure like a meta path *User-Movie-Actor-Movie-User* encodes the semantics interpretation of “movies starring the same actor are highly likely to be rated by users”. Without considering rich context information, disentangled learning fails to offer fine-grained interpretability of item attributes (e.g., *actor*) for recommendation.

Therefore, in this work, we propose to enhance user intent disentangled learning with heterogeneous information, which however is not trivial. The heterogeneous information is complicated and consists of various types of data (e.g., user/item aspects), which is usually grouped by its attributes (e.g., *gender*, *actor*). These aspect groups are frequently presented with skewed distributions that can bias the user preference and prediction score. The skewed distribution is attributed to missing values of aspects, i.e., the number of non-missing aspects is not evenly distributed in observational dataset. An empirical study conducted on Douban Movie dataset can validate this claim by Fig. 1: unobserved *Director* aspect accounts for 19.7% of items compared to *Actor* accounting for 7.6%. That is, the skewed distribution of aspects can easily bias the prediction model towards the majority group, even though their items have the same matching level (see example in Fig. 1).

In this work, we attempt to tackle the challenge from a novel causal perspective, to develop an unbiased and interpretable disentangled approach on heterogeneous information, namely, *CaDSI* (Causal Disentanglement for Semantics-Aware Intent Learning). To make users' intents semantic-aware, we propose a pre-trained model as the first stage to leverage multiple item facets of heterogeneous information. As a second stage, besides considering items directly interacted by the user, the higher-order interacted items via meta paths are exploited to disentangle user intents in a robust manner. Finally, the pre-trained model together with the disentangled learning are subsequently fine-tuned by the causal intervention. Thanks to the development of causal inference, the final module is designed to adopt a causal intervention mechanism to eliminate the bias introduced by heterogeneous information. With these

stages, our method can guide the unbiased and semantic-aware representations disentangling user intents for the robust recommendation. The key contributions of this work are fourfold:

- Fundamentally different from previous works, *CaDSI* is the first method that can disentangle the unbiased user intents from a causal perspective, in the meanwhile endow each user intent with specific semantics under the disentanglement learning task.
- We design a novel causal graph for the qualitative analysis of causal relationships in recommendations, based on which a pre-trained model is developed. With heterogeneous information, the pre-trained model can semantically account for the item aspects influence towards the user intent.
- To eliminate confounding bias stemmed from heterogeneous information, we perform the causal intervention on user representations and refine the pre-trained model for unbiased user intent learning.
- We conduct extensive experiments to show that our *CaDSI* method outperforms state-of-the-art methods. The interpretability of our *CaDSI* is also validated by our empirical study.

2 PRELIMINARY AND PROBLEM FORMULATION

This section will first introduce important concepts used in our approach and the formal definition of our problem. Next, we present causal relations underlying the data generation mechanism of recommendations with access to heterogeneous information. Following this, we prove the existence of the confounder in the heterogeneous information and discuss the consequences for ignoring the bias bought by the confounder. We finally introduce the causal intervention and discuss how intervention via back-door adjustment can control the bias from a causal perspective.

2.1 Notions and Problem Definition

We formulate our task as disentangling and interpreting users' intent based on Heterogeneous Information Network (HIN). The important concepts of HIN and the formal definition of our problem are given as follows.

Definition 1 (Heterogeneous Information Network). A Heterogeneous Information Network (HIN) is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with a node type mapping function: $\phi: \mathcal{V} \rightarrow \mathcal{A}$ and an edge type mapping function: $\psi: \mathcal{E} \rightarrow \mathcal{R}$, where $\mathcal{A}$ and $\mathcal{R}$ are the node type set and edge type set of $\mathcal{G}$, respectively. Each node $v \in \mathcal{V}$ and edge $e \in \mathcal{E}$ in a HIN belongs to one particular type with $\phi(v) \in \mathcal{A}$ and $\psi(e) \in \mathcal{R}$, where $|\mathcal{A}| + |\mathcal{R}| > 2$ (i.e., heterogeneity).

Definition 2 (Meta Path). Meta path $\mathbf{p}$ is a path defined on the network schema $T_{\mathcal{G}} = (\mathcal{A}, \mathcal{R})$, and is denoted in the form of

$$\mathbf{p} \triangleq (\mathcal{A}_1 \xrightarrow{\mathcal{R}_1} \mathcal{A}_2 \xrightarrow{\mathcal{R}_2} \dots \xrightarrow{\mathcal{R}_l} \mathcal{A}_{o+1})$$

which defines a composite $\mathcal{R} = \mathcal{R}_1 \mathcal{R}_2 \dots \mathcal{R}_l$ between type $\mathcal{A}_1$ and $\mathcal{A}_{o+1}$. For simplicity, we use node type names to denote the meta path if no multiple relations exist between type pairs, as $\mathbf{p} = (\mathcal{A}_1 \mathcal{A}_2 \dots \mathcal{A}_{o+1})$. Commonly, a HIN contains multiple meta paths, the meta path set is defined as $\mathcal{P}$ where each $\mathbf{p} \in \mathcal{P}$.

TABLE 1
Key Notations and Descriptions

Notation	Description
$\mathcal{G}$	Heterogeneous Information Network (HIN)
$\mathcal{V}$	node set of HIN
$\mathcal{A}$	node type set of HIN
$\mathcal{P}$	meta paths set of HIN
$\mathbf{p}$	a meta path in $\mathcal{P}$
$\mathbf{y} \in \mathbb{R}^{m \times n}$	user item interaction matrix
$\hat{y}_{ui}$	predicted interaction likelihood of user u on item i
$\mathbf{c}_u$	semantic-aware embedding for user u
$\mathbf{c}_i$	semantic-aware embedding for item i
$\mathbf{c}_a$	context embedding for aspect a
k	user intent number
l	iteration number of graph disentangling module
L	L -th layer of graph disentangling module
$S_k(u, i)$	intent score of u and i on intent k
$\mathbf{u}^u$	intent-aware embedding for u
$\mathbf{i}^i$	intent-aware embedding for item i

Based on these important concepts, we collect the key notations in Table 1 and formulate the problem to be solved as follows.

Definition 3 (Problem Definition). Given user and item sets, we define an interaction matrix $\mathbf{y} \in \mathbb{R}^{m \times n}$ where entry $y_{ui} = 1$ indicates a user u in user set interacts with an item i in item set, otherwise $y_{ui} = 0$. We also have additional contextual information about users and items, e.g., social relationships between users or item brands and categories, absorbing in $\mathcal{G}$ of Definition 1. Thus, we aim to learn the prediction function P parameterized by Θ , such that $\hat{y}_{ui} = P(u, i | \mathbf{y}, \mathcal{G}; \Theta)$, where $\hat{y}_{ui}$ denotes the probability that user u will engage with item i conditional on the given $\mathbf{y}$ and $\mathcal{G}$.

2.2 A Causal View on Recommendation

2.2.1 Structural Causal Model

We consider the structural causal model (SCM) [15] based on causality to reveal the true causal relations in recommendations. The basic idea of our proposed approach is to disentangle and interpret users' intent based on heterogeneous information. From a causal perspective, Fig. 2a demonstrates the illustrative causal graph that offers an interpretable representation for disentangled recommender system, which consists of four variables including $\{U, C, E, Y\}$. In particular, as a directed acyclic graph, it can describe the generation mechanism of recommendation results and guide the design of recommendation methods. In the following, we explain the rationality of this causal graph at a higher level.

- C as a confounder is the item aspects (e.g., *actor*) acquired from HINs. The representation of C can be learned by a pre-trained model from a HIN, which retains semantics information of item aspects.
- U denotes the user representation which essentially reveals k user intents. U is presented in the form of k chunked intent representation, where each chunk reveals a piece of user intent, such as the user's preference towards items' brand.

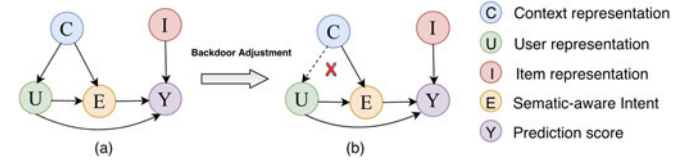


Fig. 2. Causal disentanglement model for recommendation. We apply a backdoor adjustment to remove the effect of confounder C for U , as indicated by the red cross.

- I is the item representation and each I denotes the embedding of one item attribute (e.g., *actor*).
- E is the semantic-aware intent representation generated by the context information from C and the user representation U . E retains the information of the user intent towards different item aspects.
- $Y \in [0, 1]$ is the recommendation probability of the user-item pair.

The directed edge represents the causal relation between two variables, in particular, the rationality of causal relations can be explained as follows.

- $C \rightarrow U$: The prior knowledge C of item aspects affects user representation U , which is reflected by the fact that users prefer items who have particular attributes (e.g., *actor*).
- $(C, U) \rightarrow E$: Item context C and user representation U consist of the semantic-aware user intent representation.
- $I \rightarrow Y$: Item representation by I affects the recommendation probability Y .
- $U \rightarrow Y$: User's preference represented by U affects the recommendation probability Y .
- $U \rightarrow E \rightarrow Y$: The recommendation probability of the item could be high if the user shows interest in the context of the item, e.g., the item type rather than the item. For example, items whose type is "Lipstick" are more likely to be purchased by the user u whose gender is female.

2.2.2 Adjusting Confounding Bias Via Intervention

From this causal graph, the semantic knowledge C is a confounder between user representation U and recommendation outcome Y , since C is the common cause of U and Y by definitions in causal theory. The presence of confounder C leads to the spurious correlation between U and Y if we ignore to account its causal effect into modeling, which is equal to the estimation of $P(Y|U)$. In semantic knowledge-aware recommendation, the confounder C (i.e., semantic knowledge) makes $P(Y|U)$ biased towards items that have dominant item aspects. For example, as illustrated in Fig. 1, we expect that the rating prediction of RWM is caused by both of the three item attributes, but not only the dominant *director* which has a majority attribute popularity (i.e., the majority group). In the language of causal inference, the correlation $P(Y|U)$ fails to capture the true causality between U and Y , since the prediction of Y conditions not only on U , but also the spurious correlations of (1) $C \rightarrow U \rightarrow Y$, i.e., prior knowledge C determines the prediction likelihood through user representation U . For example, the undesirable and low-quality items in specific aspect groups will not

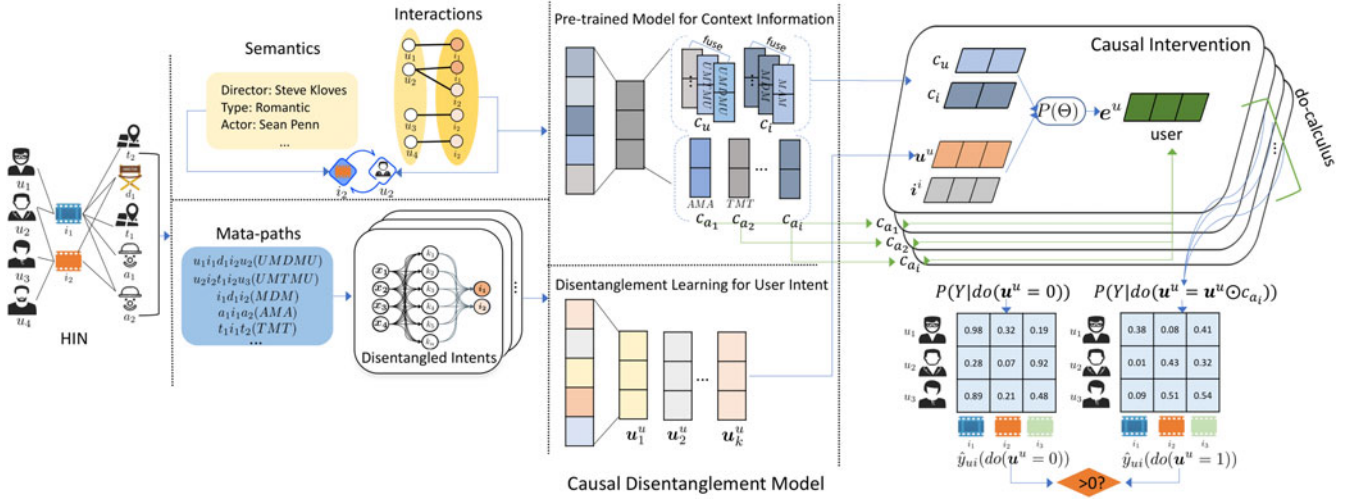


Fig. 3. Overview of the proposed CaDSI. CaDSI takes HIN as the input, and passes the causal disentanglement model for learning semantic-aware intent representations (cf. Section 3.3); then uses the causal intervention (cf. Section 3.4) for controlling the confounding bias.

attract users' intent. (2) $C \rightarrow E \rightarrow Y$. i.e., the semantic-aware user intent representation E derived from C affects the prediction Y . Once users' future interest in item aspect groups changes (i.e., user interest drift), the recommendations will be biased.

To pursue the true causality between U and Y , we should propose a causal intervention method $P(Y|do(U))$ to remove the confounding bias from C . The $do(\cdot)$ operation [15] is to forcibly and externally assign a certain value to the variable U , which can be intuitively seen as removing the edge $C \rightarrow U$ and blocking the effect of C on U (as shown in Fig. 2b). As a result, the prediction likelihood can be independent of its causes, so as to generate robust recommendations that is free from the confounding bias.

3 OUR METHOD

In this section, we first introduce our motivations and the overall architecture of the proposed model, which includes *Causal Disentanglement Model* and *Causal Intervention* as shown in Fig. 3. We then present the details of each component and how they are applied to Top- K recommendation.

3.1 Pre-Trained Model for Learning Context Information

The pre-trained model for learning context representation C is a key component in the causal disentanglement model, which aims to leverage a given HIN to construct semantic-aware representations for users, items and aspects directly. Specifically, given a HIN $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and its corresponding meta paths set $\mathcal{P}$, we aim to learn semantic-aware representations (a.k.a., embedding) c_u for each user node $u \in \mathcal{V}$, c_i for each item node $i \in \mathcal{V}$ and c_a for a specific type of aspect a (e.g., *Actor*).

Towards this, a *Heterogeneous Skip-Gram with Meta Path Based Random Walks* is designed to output a set of multinomial distributions, while each distribution corresponding to one type of node (i.e., *User*, *Item* and *Aspect* type); The *Meta Path Based Random Walks* is used to generate node sequences that capture complex semantics under a given HIN, while

Heterogeneous Skip-Gram takes the generated node sequences as inputs and catches heterogeneous neighbors of a node to output the semantic-aware representations. Finally, the semantic-aware representations for users, items and aspects are given by aggregating every node representation under different meta paths by an *Embedding Fusion*.

3.1.1 Meta Path Based Random Walks

The *Meta Path Based Random Walks* [16] generates node sequences that can capture both the semantic and structural correlations between different types of nodes. The basic idea is to put random walkers [17] in a HIN to generate paths that constitute multiple types of nodes. Specifically, given $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \phi, \psi)$, the node sequence $\mathbf{n}_p = \{v_1, \dots, v_{i+1}\}$ under a specific meta path $\mathbf{p} = (A_1 A_2 \dots A_{o+1})$ is generated according to the following distribution:

$$P(v_{i+1} | v_i, \mathbf{p}) = \begin{cases} \frac{1}{|\mathcal{N}_{v_i}^{(A_{o+1})}|}, & (v_i, v_{i+1}) \in \mathcal{E} \text{ and } \phi(v_{i+1}) = A_{o+1} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $\mathcal{N}_{v_i}^{(A_{o+1})}$ is the first-order neighbor set for node v_i whose type is A_{o+1} . The v_{i+1} is the $i+1$ -th node whose type is A_{o+1} , and v_i is the i -th node in the walk which belongs to type A_o . By regulating $v_i \in A_o$ while $v_{i+1} \in A_{o+1}$, the node types sampled by random walkers are conditioned on the pre-defined meta path $\mathbf{p}$.

Following the pre-defined meta paths in Table 3, by performing the *Meta Path Based Random Walks* on each meta path $\mathbf{p} \in \mathcal{P}$, we can obtain node sequences set for all meta paths as $\mathbf{n}^p = \{\mathbf{n}_1, \dots, \mathbf{n}_{|p|}\}$. As we care about semantic-aware embeddings for users, items and aspects, we select meta paths starting with *user*, *item* or *Aspect* type and reorganize its corresponding node sequences into *user* type-specific set $\mathbf{n}(U)$ as $\mathbf{n}(U) = \{\mathbf{n}_1, \dots, \mathbf{n}_m\}$, *item* type-specific set $\mathbf{n}(I)$ as $\mathbf{n}(I) = \{\mathbf{n}_1, \dots, \mathbf{n}_n\}$ and *aspect* type-specific set $\mathbf{n}(A) = \{\mathbf{n}_1, \dots, \mathbf{n}_h\}$. We then use *Heterogeneous Skip-Gram* to generate semantic-aware embeddings of node sequences in $\mathbf{n}(U)$, $\mathbf{n}(I)$ and $\mathbf{n}(A)$.

3.1.2 Heterogeneous Skip-Gram

Given $\mathbf{n}(U)$, $\mathbf{n}(I)$ and $\mathbf{n}(A)$ generated from Eq. (1), we aim to leverage *Heterogeneous Skip-Gram* [16] to learn semantic-aware user representation c_u for sequence $\mathbf{n}_u$ in $\mathbf{n}(U)$. Analogously, c_i is the semantic-aware item representation for sequence $\mathbf{n}_i$ in $\mathbf{n}(I)$ while aspect representation c_a is learned for sequence $\mathbf{n}_a$ in $\mathbf{n}(A)$.

Specifically, a *Heterogeneous Skip-Gram* is designed to learn node representations by aggregating node neighbors in node sequences [16]. Given each user node sequence $\mathbf{n}_u$ in $\mathbf{n}(U)$, the *Heterogeneous Skip-Gram* model learns the semantic-aware embedding c_u of $\mathbf{n}_u$ by maximizing the probability of having the heterogeneous neighbors $\mathcal{N}_u$ given a node u :

$$\mathcal{L}_\theta = \sum_{u \in \mathcal{V}} \sum_{u_c \in \mathcal{N}_u^{\mathcal{A}_i}} \sum_{\mathcal{A}_i \in \mathcal{A}} \left(\sigma(c_u^T c_{u_c}) \prod_{w=1}^W \sigma(c_u^T c_w); \theta \right) \quad (2)$$

where $\mathcal{N}_u^{\mathcal{A}_i}$ denotes u 's neighbors whose type is $\mathcal{A}_i$ and u_c is one node in the neighbors set $\mathcal{N}_u$. The c_u and c_{u_c} are latent vectors that correspond to the target node and context node representations of u and u_c , and $\sigma(x) = 1/(1 + \exp(-x))$. The W is a parameter that determines the number of negative examples to be drawn per a positive example. The c_w is the sampled node's representation within W negative samples and θ is the model parameters of *Heterogeneous Skip-Gram*. Finally, c_u for each node sequence $\mathbf{n}_u$ in $\mathbf{n}(U)$ are estimated by applying gradient descent algorithm [18]. Analogously, we can build semantic-aware item representation c_i and the aspect representation c_a .

3.1.3 Embedding Fusion

Since we have multiple node sequences in $\mathbf{n}(U)$, $\mathbf{n}(I)$ and $\mathbf{n}(A)$, we should perform embedding fusion to aggregate every representations c_u , c_i and c_a of sequences in $\mathbf{n}(U)$, $\mathbf{n}(I)$ and $\mathbf{n}(A)$ into an uniform manner. The reason why we fuse the individual embedding of each node sequence is straightforward. First, in a recommender system, the optimization goal is to learn effective representations for users and items. Hence, it requires a principled fusion way to transform node embeddings w.r.t. different meta paths relating *user* type or *item* type into a suitable form for later recommendation tasks. Second, the context information across meta paths starting with an aspect type should be further arranged into an uniform embedding space, representing one piece of semantic factors.

The embedding fusion is implemented as a liner combination function defined as follows:

$$\begin{aligned} c_u &\leftarrow \frac{1}{|c_u(U)|} \sum_{j=1}^{|c_u(U)|} (\mathbf{M} \cdot c_u^j + b) \\ c_i &\leftarrow \frac{1}{|c_i(I)|} \sum_{j=1}^{|c_i(I)|} (\mathbf{M} \cdot c_i^j + b) \\ c_a &\leftarrow \frac{1}{|c_a(A)|} \sum_{j=1}^{|c_a(A)|} (\mathbf{M} \cdot c_a^j + b) \end{aligned} \quad (3)$$

where $c_u(U)$ is the user node representation set that absorbs the node representations of user u and $c_u(U) = \{c_u^1, \dots, c_u^m\}$.

Correspondingly, the item and aspect node representation sets $c_i(I) = \{c_i^1, \dots, c_i^n\}$ and $c_a(A) = \{c_a^1, \dots, c_a^h\}$ are established for item i and aspect a . $\mathbf{M}$ is a linear combination transformation matrix [19] and b is the error term. Through Eq. (3), c_u , c_i and c_a can be learned as final semantic-aware representations for a user u and an item i and context representation for aspect a , respectively.

3.2 Disentanglement Learning for User Intent

Inspired by recent achievements on GNNs [12], [13], [20], [21], we propose a GNN-based disentangling module to learn user representations that can essentially reveal k user intents. Specifically, the L -layer disentangling module exploits the high-order connectivities among the user-item interaction graph and initializes intent-aware embeddings by separating each user/item embedding into k chunks. Then, an interactive update rule that computes the importance scores of intent-aware user-item interactions is designed to refine intent-aware embeddings, so as to disentangle the holistic interaction graph into k intent-aware sub-graphs. Thereafter, each intent-aware embedding chunk are stacked by embedding propagation in the current layer, serving as the holistic intent-aware embedding u^u and i^i for user u and item i , where each intent-aware embedding u^u and i^i is composed of k independent chunks:

$$u^u = [u_1^u, \dots, u_k^u], \quad i^i = [i_1^i, \dots, i_k^i] \quad (4)$$

Each chunked representation $u_k^u \in \mathbb{R}^{\frac{d}{k}}$ and $i_k^i \in \mathbb{R}^{\frac{d}{k}}$ is built upon the intent-aware interactions between user u and its preferred items under intent i . We ultimately sum up intent-aware representations at each intent k of all L layers, the final layer outputs the k -th chunked intent-aware representations u_k^u and i_k^i :

$$u_k^u = u_k^u(1) + \dots + u_k^u(L), \quad i_k^i = i_k^i(1) + \dots + i_k^i(L) \quad (5)$$

The detailed operations are given as follows.

3.2.1 Initialization

As ID embedding captures intrinsic characteristics of users, we separate the ID embedding x of user u into k chunks and associate each chunk with a latent intent. Then the x serves as the initialization of u^u in Eq. (4):

$$x = [x_1, x_2, \dots, x_k] \quad (6)$$

Thereafter, we initialize the importance score $\mathbf{S}_k(u, i)$. The $\mathbf{S}_k(u, i)$ is the importance score of the interaction between u and i concerning the k -th intent. Such $\mathbf{S}_k(u, i)$ can be seen as the indicator of whether u should interact with i under a specific intent k . Thus, by learn $\mathbf{S}_k(u, i)$ for aspect $k \in \{1, \dots, k\}$, we can construct an intent-aware sub-graph for intent k . Such that the embedding propagation can output intent-aware representations at intent k for all users based on the corresponding intent-aware sub-graph. We uniformly initialize importance score $\mathbf{S}_k(u, i)$ as $\mathbf{S}_k(u, i) = \frac{1}{k}$ which presumes the equal contributions of intents at the start of modeling.

3.2.2 Iterative Update Rule

An iterative update rule is then designed to update the importance score $\mathbf{S}_k(u, i)$ within l iterations, so as to disentangle intent sub-graphs to refine each intent-aware embedding chunk $\mathbf{x}_k$ in Eq. (6). Note that $\mathbf{x}_k \in \mathbf{x}$ in Eq. (6) serves as the initialized representation chunk of $\mathbf{u}_k^u \in \mathbf{u}^u$ in Eq. (4) and is used to memorize the updated value during iteration. The final $\mathbf{x}_k$ is assigned to each $\mathbf{u}_k^u$ as the final intent-aware representation chunk of user u .

In particular, we set l iterations in the interactive update. At each iteration, for the target interaction (u, i) , we firstly normalize the score vector $\mathbf{S}_k(u, i) \mid \forall k \in \{1, \dots, k\}$ over all intents into $\tilde{\mathbf{S}}_k$ via a softmax function:

$$\tilde{\mathbf{S}}_k^l(u, i) = \frac{\exp(\mathbf{S}_k^l(u, i))}{\sum_{k'=1}^k \exp(\mathbf{S}_{k'}^l(u, i))} \quad (7)$$

which is capable of illustrating which intent should get more attention in the user-item interaction (u, i) .

We then learn the representation $\mathbf{x}_k$ of the intent-aware graph whose adjacency matrix is $\tilde{\mathbf{S}}_k$ via embedding propagation. The weighted sum aggregator of the embedding propagation is defined as:

$$\mathbf{x}_k^l = \sum_{i \in \mathcal{N}_u} \frac{\tilde{\mathbf{S}}_k^l(u, i)}{\sqrt{D_k^l(u) \cdot D_k^l(i)}} \cdot \mathbf{i}_k^i \quad (8)$$

where $\mathbf{x}_k^l$ is $\mathbf{u}_k^u$'s temporary representation that memorizes information acquired from $\mathcal{N}_u$ at the l -th iteration. The $D_k^l(u) = \sum_{i' \in \mathcal{N}_u} \tilde{\mathbf{S}}_k^l(u, i')$ and $D_k^l(i) = \sum_{u' \in \mathcal{N}_i} \tilde{\mathbf{S}}_k^l(u', i)$ are the degrees of user u and item i , respectively.

Then we interactively update intent-aware graphs. Intuitively, historical items of users driven by the same intent tend to have similar chunked representations. This goal can be achieved by encouraging users and items among the same intent sub-graph to have stronger relationships. We hence iteratively update the interaction importance score $\mathbf{S}_k^l(u, i)$ to strengthen the degree between the centroid u and its neighbor i under intent k , as follows:

$$\mathbf{S}_k^{l+1}(u, i) = \mathbf{S}_k^l(u, i) + \mathbf{x}_k^{l\top} \tanh(\mathbf{i}_k^i) \quad (9)$$

where $\mathbf{x}_k^{l\top} \tanh(\mathbf{i}_k^i)$ considers the affinity between $\mathbf{x}_k^l$ and $\mathbf{i}_k^i$, and $\tanh$ [22] is a nonlinear activation function that increases the representation ability of model.

After l iterations, $\mathbf{u}^u(1) = \mathbf{x}^l(1)$ for user u is obtained in the first layer, where each chunked representation $\mathbf{u}_k^u(1) = \mathbf{x}_k^l(1)$ denotes the chunked embedding of $\mathbf{u}^u(1)$ on the intent k corresponds to the k -th dimension of Eq. (4).

3.2.3 Layer Combination

To explore high-order connectivities between users and items, we recursively formulate the representation of L -th layer as:

$$\mathbf{u}_k^u(L) = g(\mathbf{x}_k^l(L-1), \{\mathbf{i}_k^i(L-1) \mid i \in \mathcal{N}_u\}) \quad (10)$$

where $\mathbf{u}_k^u(L)$ and $\mathbf{i}_k^i(L)$ are the representations of user u and item i on the k -th intent at layer L . The $\mathbf{x}_k^l(L-1)$ is the chunked embedding of k -th intent at $L-1$ -th layer of $\mathbf{u}_k^u(L-1)$. Note that $g(\cdot)$ is a fully connection layer

memorizes the information propagated from the $(L-1)$ -order neighbors of u .

Finally, after L layers, we sum up intent-aware representations at different layers as the final representation of the k -th chunk of Eq. (4), as $\mathbf{u}_k^u = \mathbf{u}_k^u(1) + \dots + \mathbf{u}_k^u(L)$, where $\mathbf{u}_k^u$ donates the intent-aware representation for user u at intent k . Analogously, we can establish the final intent-aware embedding chunk $\mathbf{i}_k^i$ following the definition in Eq. (5).

3.3 Semantic-Aware Intent Learning

Having obtained the intent-aware embeddings $\mathbf{u}^u, \mathbf{i}^i$ from *Disentanglement learning for User Intent*, our method takes semantics factors $\mathbf{c}_u$ and $\mathbf{c}_i$ from *Pre-trained Model for Context Information* as the auxiliary inputs for the semantic-aware recommendation. Specifically, we design an operator to instantiate a semantic-aware intent representation $\mathbf{e}$ that incorporates intent-aware embeddings and semantics factors. The intent representation $\mathbf{u}^u$ is updating through training together with $\mathbf{e}$ to capture the effect of semantics factors. The learned $\mathbf{u}^u$ retains the information from semantic factors and can be easily plugged into the backdoor adjustment to alleviate bias. The second-order Factorization Machine (FM) [23] operator for instantiating $\mathbf{e}$ is defined as:

$$\mathbf{e} = \sum_{a=1}^d \sum_{b=1}^d \mathbf{u}_a^u \mathbf{c}_{i_a} \odot \mathbf{c}_{u_b} \mathbf{i}_b^i \quad (11)$$

where $\odot$ denotes the element-wise product that is used for capturing interactions between intent representations and semantics factors.

Next, the semantic-aware intent representation $\mathbf{e}$ can be incorporated into recommender models as one additional user representation. Formally, we use the collaborative filtering to calculate the prediction score $\hat{\mathbf{y}}_{ui}$ given user and item ID representations, as follows:

$$\hat{\mathbf{y}}_{ui} = f(\mathbf{u}, \mathbf{i}, \mathbf{e}) = \delta \mathbf{u}^\top \mathbf{i} + (1 - \delta) \mathbf{e}^\top \mathbf{i} \quad (12)$$

where $\mathbf{u}$ and $\mathbf{i}$ are the ID embeddings given by Multi-One-Hot [24], and δ is the coefficient that describes how much each component contributes to the prediction score.

Then we use the pairwise BPR loss [25] to optimize the model parameters Θ . Specifically, BPR loss encourages the prediction likelihood of a user's historical items to be higher than those of unobserved items:

$$\mathcal{L}_{\text{BPR}} = \sum_{(u, i, j) \in O} -\ln \sigma(\hat{\mathbf{y}}_{ui} - \hat{\mathbf{y}}_{uj}) + \lambda \|\Theta\|_2^2 \quad (13)$$

where $\mathbf{u}, \mathbf{i}$ and $\mathbf{j}$ are the ID embeddings of user u , item i and item j , $O = \{(u, i, j) \mid (u, i) \in O^+, (u, j) \in O^-\}$ denotes the training dataset involving the observed interactions O^+ and unobserved counterparts O^- . The $\sigma(\cdot)$ is sigmoid function and λ is the coefficient controlling regularization.

3.4 Causal Intervention for Debiasing

The context information as a confounder negatively affects both user representations and prediction scores. To make context information beneficial for semantic-aware intent learning, we resort to a causal inference technique, namely, backdoor adjustment [15], to adjust the user representations

in the disentangled causal model. By doing so, we aim to produce unbiased user representations and apply the modified ones to the recommendation task.

3.4.1 Backdoor Adjustment

Note that previous recommenders build predictive models $P(Y|U)$ from the passively collected interaction dataset, which neglect the effect of confounder C and lead to spurious correlations between users and items. The spurious correlation is harmful since the items in the majority group are likely to dominate the recommendation list and narrow down the user interests.

According to the theory of backdoor adjustment [15], we aim to remove the harmful effect of context information C on user representation U . Instead of $P(Y|U)$, we formulate the predictive model as $P(Y|do(U = u))$ to account for the effect of confounder. Based on the graph in Fig. 2, we first need to formulate the causal effect between variables by causal intervention, which is denoted by $do(\cdot)$ operation [15]. The operation $do(U = u)$ is defined to externally assign a certain value to U . Namely, $do(U = u)$ intuitively removes the edge $C \rightarrow U$ so as to block the effect of C on U , making the value of U independent of its causes (cf. Fig. 2). By applying do operation, we can estimate the effect of C on Y by:

$$P(Y|do(U = u)) - P(Y|do(U = 0)) \quad (14)$$

where $P(Y|do(U = 0))$ denotes the null intervention, e.g., the baseline compared to $U = u$. In the physical world, $P(Y|do(U = u))$ corresponds to actively manipulating the aspects or attributes in the item.

Our implementation is inspired by two inherent properties of any Heterogeneous Network Embedding method (e.g., metapath2vec++ [16]). First, by passing a meta-path into the pre-trained context model in Eq. (3), we have the context embedding c_a for an aspect (e.g., *actor*). Aggregating context embeddings for all aspects into the aspect representation set C , we can obtain the unified aspect representation C , where each element of C , i.e., c_a , representing one semantic aspect. As C is no longer correlated with u^u by removing the edge $C \rightarrow U$, the causal intervention makes u^u has a fair opportunity to incorporate every context c_a into the prediction of $\hat{y}_{ui}$, subject to a prior $P(C = c_a)$. Second, prevailing pre-trained models use specific tasks (in our method is the semantic-aware intent learning in Section 3.3) as the objective, the representations trained from it can be considered as the distilled information u^u that waits to be adjusted by $do(U = u^u \odot C)$.

By far, we have the context information C for all aspects, where each $c_a \in C$ represents the context embedding of one aspect. We also have the refined intent representation u^u that waits to be adjusted from Eq. (13) as U . Next, we will detail the causal intervention by providing implementation for Eq. (14).

The overall backdoor adjustment is achieved through:

$$\begin{aligned} & P(Y|U, do(U = u)) - P(Y|do(U = 0)) \\ &= \sum_C (P(Y|do(U = u^u \odot C)) - P(Y|do(U = 0)))P(C = c_a) \\ &= \frac{1}{N} \sum_{i=1}^N (P(\hat{y}_{ui} | u^u \odot C) - P(\hat{y}_{ui} | u^u)) \end{aligned} \quad (15)$$

where each component in Eq. (15) is designed by:

- $C = c_a$ indicates the confounder C is set as one context embedding c_a .
- $P(Y|U, do(U = u)) = P(\hat{y}_{ui} | u^u \odot C)$. The selected context embedding C is concatenated with u^u by the element-wise product $\odot$, which serves as the adjustment value for the prediction of $\hat{y}_{ui}$.
- $P(C = c_a)$ is the prior distribution of different item aspect, by defining $P(C = c_a) = 1/N$, we can assume a uniform prior for the adjusted features, where N is the total number of aspect type.

We then utilize an inference strategy to adaptively fuse the prediction scores from the $P(\hat{y}_{ui} | u^u \odot C)$ and $P(\hat{y}_{ui} | u^u)$. Specifically, we first train the recommender models by $P(Y|do(\hat{y}_{ui} = u^u \odot C))$ and $P(Y|do(\hat{y}_{ui} = 0))$ and obtain $\hat{y}_C$ and $\hat{y}$, respectively. Then, the adjusted prediction score $\hat{y}_C$ and unadjusted prediction score $\hat{y}$ are automatically fused to regulate the impact of backdoor adjustment. We define an indicator function $\mathbf{I}_a$ that determines whether to include c_a into the user intent u^u or not.

$$\mathbf{I}_a := \begin{cases} c_a & \text{if } \tanh(\hat{y}_C - \hat{y}) > 0 \\ 1 & \text{if } \tanh(\hat{y}_C - \hat{y}) < 0 \end{cases} \quad (16)$$

where $\tanh(\hat{y}_C - \hat{y}) > 0$ indicates that the backdoor adjustment leads a positive impact on recommendation result by considering the aspect c_a . Otherwise, c_a leads a negative impact, which should be removed from the user intent representation. Based on Eq. (16), the semantic-aware user intent representation can be refined as follows:

$$e_a = u^u \odot \mathbf{I}_a \quad (17)$$

To define the unbiased loss function for the observation $\hat{y}_{ui}$, we aim to maximize the discrepancy between the final adjusted and the unadjusted representation u^u under the guidance of e_a , that is,

$$\mathcal{L}_d = \arg \min_{\theta} \sum_{(u,i,\mathbf{y}_{ui}) \in O} \left(\mathbf{y}_{ui}, f\left(u, i, \prod_a e_a\right) \right) \quad (18)$$

where u, i are the ID embeddings for user u and item i , $f(\cdot)$ is defined in Eq. (12), and $\mathbf{y}_{ui}$ is the ground-truth for the interaction between user u and item i .

3.5 Optimization

Our model ultimately has three loss functions, i.e., $\mathcal{L}_d$ of unbiased preference score estimation loss given in Eq. (18), $\mathcal{L}_\theta$ of *Heterogeneous Skip-Gram* model given in Eq. (2), and the BPR loss given in Eq. (13). To this end, the objective function of our CaDSI method could be derived as:

$$\mathcal{L} = \lambda_d \mathcal{L}_d + \lambda_\theta \mathcal{L}_\theta + \lambda_z \mathcal{L}_{BPR} + \mathcal{R}(\Omega) \quad (19)$$

where Ω represents the trainable parameters and $\mathcal{R}(\cdot)$ is a squared L_2 norm regularization term on Ω to alleviate the overfitting problem. $\lambda_d, \lambda_\theta, \lambda_z$ are trade-off hyper-parameters of the three separate loss functions respectively. During the training, we optimize the objective function in Eq. (19) via gradient descent algorithm [26].

TABLE 2
Statistics of Three Datasets

Dataset (Density)	Node	Relation A-B	Avg.Degree of A/B
MovieLens-HetRec (4.0%)	#User(U): 2,113	#U-M: 855,598	#U/M: 405.0/84.6
	#Movie(M): 10,109	#U-U: 0	#U/U: 0/0
	#Actor(A): 38,044	#M-A: 95,777	#M/A: 9.5/2.5
	#Director(D): 4,031	#M-D: 10,068	#M/D: 1.0/2.5
	#Country(C): 72	#M-C: 10,109	#M/C: 1.0/140.0
	#Genre(G): 20	#M-G: 20,670	#M/G: 2.0/1033.5
Douban Book (0.27%)	#User(U): 13,024	#U-Bo: 792,062	#U/Bo: 60.8/35.4
	#Book(Bo): 22,347	#U-U: 169,150	#U/U: 13.0/13.0
	#Group(Gr): 2,936	#U-Gr: 1,189,271	#U/Gr: 91.3/405.1
	#Author(Au): 10,805	#Bo-Au: 21,907	#Bo/Au: 1.0/2.0
	#Publisher(P): 1,815	#Bo-P: 21,773	#Bo/P: 1.0/12.0
	#Year(Y): 64	#Bo-Y: 21,192	#Bo/Y: 1.0/331.1
Douban Movie (0.63%)	#User(U): 13,367	#U-M: 1,068,278	#U-M: 79.9/84.3
	#Movie(M): 12,677	#U-U: 4,085	#U/U: 1.7/1.8
	#Group(Gr): 2,753	#U-Gr: 570,047	#U/Gr: 42.7/207.1
	#Actor(A): 6,311	#M-A: 33,587	#M-A: 2.9/5.3
	#Director(D): 2,449	#M-D: 11,276	#M/D: 1.1/4.6
	#Type(T): 38	#M-T: 27,668	#M/T: 2.2/728.1

Density of Dataset is $\#Interactions / (\#Users \cdot \#Items)$. Avg. Degree of A is $\#Relation / \#A$ and Avg. Degree of B is $\#Relation / \#B$.

4 EXPERIMENTS

To more thoroughly evaluate the proposed method, experiments are conducted to answer the following research questions:

- (RQ1) How confounding bias caused by the context information is manifested in real-world recommendation datasets?
- (RQ2) How does our model perform compared with state-of-the-art models for Top- K recommendation?
- (RQ3) How do key components in our model impact the recommendation performance? (i.e., disentanglement learning task, causal intervention)? How do hyper-parameters in our model impact recommendation performance?
- (RQ4) How does our model interpret user intents for recommendations?

We first present the experimental settings for good reproducibility, the answer the four research questions above.

4.1 Experimental Settings

4.1.1 Datasets

We evaluate our model on three publicly accessible datasets for the Top- K recommendation. The statistics of the datasets are summarized in Table 2 and the selected meta paths for all data sets are shown in Table 3. To ensure the quality of all the datasets, we use the core settings, i.e., we transform explicit ratings into implicit data, where each interaction between users and items is marked as 0 or 1 indicating whether the user has rated the item or not; we retaining users and items with at least five interactions and each user has at least five friends for both of the datasets. In the training phase, each observed user-item interaction is treated as a positive instance. We use negative sampling to randomly sample an unobserved item and pair it with the user as a negative instance.

4.1.2 Baselines

To demonstrate the effectiveness, we compare our model with four classes of methods: (I) conventional entangled CF methods; (II) graph-based entangled methods; (III) HIN enhanced entangled methods, which model user-item interaction with rich context information in a HIN; (IV) disentangled methods, which disentangle user intents or item aspects with different mechanisms; (V) causal-based recommendation methods.

- NeuMF [27] (I): This method combines deep neural networks with Matrix Factorization (MF) method for modeling user-item interactions.
- GC-MC [20] (II): The method organizes user behaviors as a graph, and employs one Graph Convolution Network (GCN) encoder to generate representations based on first-order connectivity.
- NGCF [21] (II): This adopts three Graph Neural Network(GNN) layers to model at most third-order connectivity on the user-item interaction graph.
- LightGCN [13] (II): This is a state-of-the-art graph-based recommendation method that learns user/

TABLE 3
The Selected Meta Paths for Three Datasets Used in Experiments

Dataset	Meta path Schemes
MovieLens-HetRec	UMU, UMAMU, UMDMU, UMCMU, UMGMU
Douban Book	MUM, MAM, MDM, MCM, MGM UBoU, UBoAuBoU, UBoPBoU, UBoYBoU, UBoAuBoU
Douban Movie	BoUBo, BoPBo, BoYBo, BoAuBo UMU, UMDMU, UMAMU, UMTMU MUM, MAM, MDM, MTM

item embeddings by linearly propagating them with neighbors aggregation in the GCN component.

- IF-BPR [28] (III): This method leverages meta path-based social relations derived from a HIN, and proposes a social recommender that can capture the similarity of users.
- MCRec [29] (III): This method leverages meta path-based context with co-attention mechanism for Top- K recommendation.
- NeuACF [30] (IV): This method disentangles multiple aspects of users and items with a deep neural network for HIN-enhanced recommendations.
- MacridVAE [8] (IV): This method disentangle user intents behind user behaviors, assuming that the co-existence of macro and micro latent factors affect user behaviors.
- DGCF [31] (IV): This is a state-of-the-art CF-based disentangled recommendation method, which disentangles latent factors of user intents by the neighbor routing and embedding propagation.
- DICE [11] (V): This is a state-of-the-art causal-based recommendation method, which aims at disentangling users' interest by controlling the conformity bias using causal embedding.

4.1.3 Evaluation Metrics

We adopt two popular metrics: Recall@ K and Normalized Discounted Cumulative Gain(NDCG)@ K to evaluate the Top- K recommendation performance of our model. The K is set as 20 by default. In the inference phase, we view the historical items of a user in the test set as the positive, and evaluate how well these items are ranked higher than all unobserved ones. The average results w.r.t. the metrics over all users are reported.

4.1.4 Parameter Settings

We implement all baseline models and our proposed CaDSI¹ model on a Linux server with Tesla P100 PCI-E 16GB GPU. For a fair comparison, datasets for implementing all models are split as train/test/validate set with a proportion of 80%/10%/10% of the dataset, while we optimize all models with Adam [26]. For a fair comparison, a grid search is conducted to choose the optimal parameter settings, e.g., dimension of user/item latent vector k_{MF} for matrix factorization-based models and dimension of embedding vector d for neural network-based models. The embedding size is initialized with the Xavier [32] and searched in {8, 16, 32, 64, 128, 256}. The batch size and learning rate are searched in {32, 64, 128, 512, 1024} and {0.0005, 0.001, 0.005, 0.01, 0.05, 0.1}, respectively. The maximum epoch N_{epoch} is set as 2000 and an early stopping strategy is performed. Moreover, we employ three hidden layers for the neural components of GC-MC NGCF, LightGCN, MCRec, NeuACF, MacridVAE and DGCF. The hyperparameter specifications of CaDSI are set as: latent intents number k as 4, iteration number of disentangling module l as 2, model depth of disentangling module L as 2, iteration number of

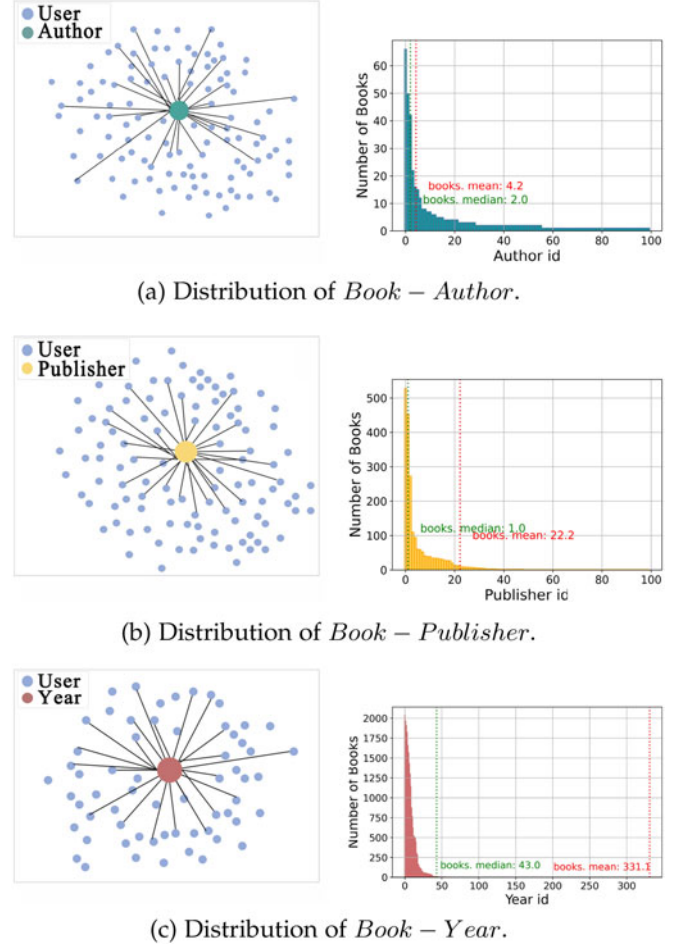


Fig. 4. The distributions of *Book – Author*, *Book – Publisher* and *Book – Year* of Douban Book dataset.

causal intervention n as 140, and their influences are reported in Section 4.4.

4.2 Understanding Confounders (RQ1)

We initially conduct an experiment to understand to what extent the confounding bias exists in meta paths of real-world recommendation datasets. To this end, we aim to investigate the distribution of nodes among the same meta path. Intuitively, an unbiased HIN-based recommendation method should expect that, for a specific attribute, each user/item should hold an equal number of this attribute (i.e., interactions between nodes and attributes are likely to be evenly distributed). Thus, we investigate the confounding bias by analyzing the statistics of node-attribute interactions of meta paths in Douban Book. We randomly sample $n = 100$ books from Douban Book and extract their *Author*, *Publisher*, and *Year* attributes. By counting the connections between books and their attributes whose type belongs to *Author*, *Publisher*, and *Year*, respectively, we have the statistical results shown in Fig. 4. The connected graphs in the left part of Fig. 4 depict whether the book i connected with the selected attribute. The figures in the right part of Fig. 4 show the distributions of connected book numbers by a certain attribute, where the y -axis denotes the total amount of the connected books.

1. Our code is currently shared on Github: <https://github.com/AlfredYuiGood/CaDSI-Implementation-Code>

TABLE 4

Overall Performance Comparison: Bold Numbers are the Improvement Percentages; the Best Results are Marked With *, Strongest Baselines are Marked With Underline

	MovieLens-HetRec				Douban Book				Douban Movie			
	Recall@20	NDCG@20	Recall@40	NDCG@40	Recall@20	NDCG@20	Recall@40	NDCG@40	Recall@20	NDCG@20	Recall@40	NDCG@40
NeuMF	0.0434	0.0557	0.0665	<u>0.0709</u>	0.0339	0.0391	0.0641	0.0682	0.0460	0.0417	0.0708	0.0611
GC-MC	0.0336	0.0404	0.0653	<u>0.0584</u>	0.0458	0.0402	0.0675	0.0643	0.0448	0.0461	0.0602	0.0622
NGCF	0.0365	0.0508	0.0699	0.0615	0.0252	0.0301	0.0707	0.0691	0.0475	0.0498	0.0689	<u>0.0642</u>
LightGCN	0.0466	0.0155	0.0615	0.0498	0.0201	0.0225	0.0531	0.0568	0.0294	0.0331	0.0499	0.0578
IF-BPR	0.0546	0.0510	0.0727	0.0689	0.0396	0.0463	0.0628	0.0601	0.0483	0.0501	0.0652	0.0603
MCR	0.0352	0.0195	0.0680	0.0677	0.0165	0.0294	0.0481	0.0507	0.0281	0.0336	0.0618	0.0629
NeuACF	0.0236	0.0308	0.0556	0.0684	0.0298	0.0201	0.0601	0.0579	0.0351	0.0438	0.0571	0.0623
MacridVAE	0.0454	0.0290	0.0661	0.0592	0.0309	0.0425	0.0691	0.0645	0.0489	0.0441	0.0729	0.0616
DGCF	0.0229	<u>0.0589</u>	0.0532	0.0708	0.0431	0.0502	0.0649	0.0663	0.0416	<u>0.0527</u>	0.0702	0.0628
DICE	<u>0.0549</u>	0.0499	<u>0.0740</u>	0.0703	<u>0.0577</u>	<u>0.0608</u>	<u>0.0820</u>	<u>0.0799</u>	<u>0.0513</u>	0.0389	<u>0.0811</u>	0.0636
Our model	0.0678*	0.0659*	0.0765*	0.0736*	0.0708*	0.0722*	0.1466*	0.1194*	0.0583*	0.0547*	0.0918*	0.0647*
%improv.	23.5%	11.9%	3.4%	3.8%	22.7%	18.8%	78.8%	49.4%	13.6%	3.8%	13.2%	0.8%

Apparently, attributes and books exhibit an unevenly distribution regarding their interactions: the attributes in dataset are partially observed, leaving a larger number of attributes to be unobserved. For example, for *Book-Author* meta path, there are a lot of books that do not connect with any node whose type is *Author*, which means lots of author attributes of books are missing. In conventional recommendation methods, the missing pattern of such attributes is ignored by either regarding them as outliers and padding them with random values, or treating the missing attributes as negative feedback. Such measure would preserve the confoundings brought by node attributes, degrading the recommendation ultimately.

Another finding is that the distribution for book-attribute connection numbers is significantly skewed: it displays a long-tail phenomenon where the green vertical line separates the top 50% of connection numbers by popularity; these connections outweigh another 50% long-tail connections to the right. For instance, in Fig. 4a, authors in the *Book-Author* relation cumulatively connect with 90% more books than the long-tail authors to the right. For *Book-Publisher* meta path, ideally, *Book-Publisher* has the one-to-one relation from book to the publisher, while every publisher has published an equal number of books. However, some publishers have published at most 510 books, while more than 90% of publishers only published fewer than 10 books. Such long-tail distribution can bias the users' interest on item aspects. i.e., recommendation methods tend to recommend items with the most frequent attribute, while users can only be exposed to recommended items.

4.3 Performance Comparison (RQ2)

We compare the Top- K recommendation performance of CaDSI with ten recommendation baselines on three datasets: MovieLens-HetRec, Douban Book and Douban Movie. Table 4 demonstrates the performance comparison and we have the following observations:

- Our CaDSI consistently yields the best performance among all methods on three datasets. In particular, CaDSI improves over the strongest baselines w.r.t.

Recall@20 by 23.5%, 22.7%, 13.6%, NDCG@20 by

11.9%, 18.8%, 3.8%, Recall@40 by 3.4%, 78.8%, 13.2% and NDCG@40 by 3.8%, 49.4%, 0.8% on MovieLens-HetRec, Douban Book and Douban Movie respectively. CaDSI outperforms all baseline methods on Top- K recommendation task, which validates that the semantic-aware user intents representation can enhance the recommendation performance.

- In virtue of user-item interaction graph and meta paths, GNN-based (GC-MC, NGCF and LightGCN) and HIN-based (IF-BPR, MCR) recommendation methods can achieve better performance than conventional MF methods (NeuMF) in most cases. However, they ignore controlling the bias existing in the context information. On the contrary, our CaDSI adopts a principled causal inference way to easing such confounding bias. So it outperforms GNN-based and HIN-based recommendation methods on both of the datasets. For instance, our CaDSI outperforms the most competitive HIN-based recommender IF-BPR w.r.t. Recall@20/Recall@40 by 24.2%/5.2% and NDCG@20/NDCG@40 by 29.2%/6.8% on MovieLens-HetRec.
- By performing unbiased disentanglement via semantics context, our CaDSI can infer user's potential interests of items. However, those user interests could not be well inferred from other disentangled recommendation methods (NeuACF, MacridVAE and DGCF).
- Among the GNN-based recommenders (GC-MC, NGCF and LightGCN), HIN-based recommenders (IF-BPR, MCR) and disentangled recommenders (NeuACF, MacridVAE and DGCF), the causal-based disentangled method (DICE) serves as the strongest baseline in most cases. This justifies the effectiveness of easing the confounding bias in context information when estimating disentangled users' interests. However, DICE performs worse than our CaDSI, as it ignores rich semantics information in HIN and fails to ingest semantics aspects when disentangling user interests.

- From movie recommendation datasets, we can find that the improvements on MovieLens-HetRec is bigger than that on Douban Movie. This is reasonable since Douban Movie is much more sparser than

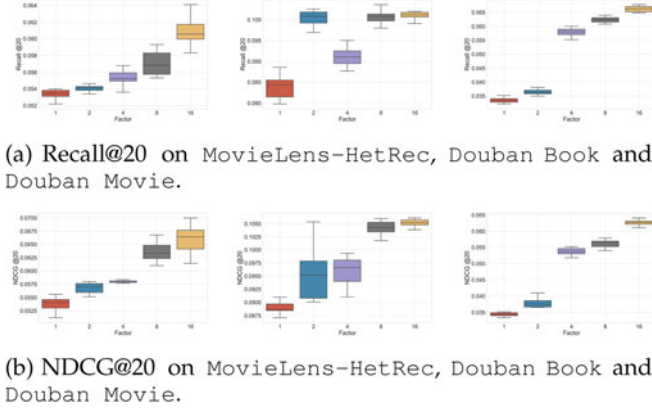


Fig. 5. The recommendation performance comparison under different latent user intent factors.

MovieLens-HetRec with density of 0.63% versus 4.0%, respectively. However, our CaDSI has better performance than all baselines on Douban Movie, because it achieves unbiased evaluation on high-order connectivity and rich semantics. This indicates that CaDSI is robust to the very sparse datasets.

4.4 Study of CaDSI (RQ3)

Ablation studies on CaDSI are also conducted to investigate rationality and effectiveness. Specifically, we first attempt to exploit how the *disentangled learning* and *causal intervention* affect our performance. Moreover, the stability of our approach's performance on the Top- K recommendation is validated as well.

We have one fixed parameter $n = 140$ (cf. Eq. (15)) which denotes the total number of causal intervention times. Three important hyperparameters k (cf. Eq. (4)), L (cf. Eq. (10)) and K (cf. Section 4.1.3) correspond to: the number of latent factors of user intents, the number of graph disentangling layers and the number of items in Top- K recommendation list, respectively. Based on the hyperparameter setup in Section 4.1.4, for all questions listed above, we vary the value of one parameter while keeping the others unchanged.

4.4.1 Effect of Disentanglement Learning

The intent number k controls the total amount of user intents considered in our model; larger k stands for more fine-grained disentangled user intents. To study the influence, we vary k in the range of $\{1, 2, 4, 8, 16\}$ and show the corresponding performance comparison on MovieLens-HetRec, Douban Book, Douban Movie in Fig. 5. We have several observations.

- Increasing the intent number from 1 to 16 can significantly enhance the performance, while CaDSI performs the worst when $k = 1$. This indicates learning the disentangled user intents is effective to capture the real user preferences towards items instead of coupling all preference together.
- The variations are diverse across different datasets. For MovieLens-HetRec and Douban Movie, the performance of CaDSI increase steadily as the k value increases from 1 to 16, while the performance drops when k is set from 2 to 4 on Douban Book. One

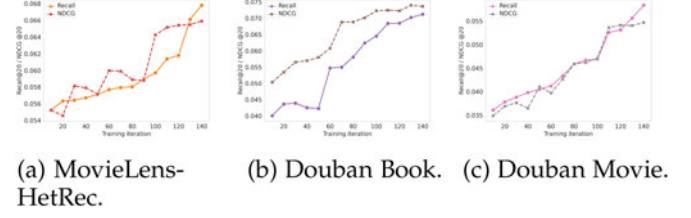


Fig. 6. Impact of causal intervention on the recommendation performance of our CaDSI along with iterations.

possible reason is that CaDSI should balance between too fine-grained disentangled intents and the adjustment from causal intervention, such balancing learning is more obvious when dataset size is larger.

4.4.2 Effect of Causal Intervention

To investigate whether CaDSI can benefit from causal intervention, we study the performance of CaDSI by varying the iterations of causal intervention. Fig. 6 summarizes the experimental results w.r.t. MovieLens-HetRec, Douban Book, Douban Movie and we have the following observations:

- Clearly, the causal intervention mechanism renders our CaDSI a better recommendation performance: more iterations of causal intervention lead to better recommendation performance before saturation on all datasets, e.g., the Recall@20 and NDCG@20 values generally increase along with training iterations in Fig. 6.
- When training iterations reach to 130, 70 and 110 for MovieLens-HetRec, Douban Book and Douban Movie, respectively, the performance becomes relatively stable. Moreover, Douban Book requires fewer iteration times than the other two datasets. Intuitively, purchasing books is a much more simple behavior than choosing movies. Thus, user intents on book aspects are less diverse, leading to a quick convergence to the optimal interventional representations.
- Some fluctuations appear in the iteration process, especially on MovieLens-HetRec dataset. The size of MovieLens-HetRec is much smaller than the other two datasets, thus leading to instability to intervention process due to the data sparsity. However, when carrying more iterations, small-size datasets such as MovieLens-HetRec can also yield satisfying results.

4.4.3 Effect of Multi-Order Connectivity

Since CaDSI is benefited from the higher-order connectivity between complex interactions and context information, we investigate how connectivity degrees affect the CaDSI. Specifically, we search the graph disentangling layer number L in the range of $\{1, 2, 3\}$, which correspond to first-order connectivity, second-order connectivity and third-order connectivity, respectively. We show the performance comparison in Table 5 and below are our observations.

- More graph disentangling layers will collect more information from multi-hop neighbors from a holistic

TABLE 5
Impact of Multi-Order Connectivity (i.e., Graph Propagation Layer Number L) on MovieLens-HetRec, Douban Book and Douban Movie

Layer number	MovieLens-HetRec		Douban Book		Douban Movie	
	Recall	NDCG	Recall	NDCG	Recall	NDCG
1	0.0505	0.0521	0.0651	0.0684	0.0583	0.0546
2	0.0672	0.0683	0.0712	0.0736	0.0596	0.0570
3	0.0611	0.0624	0.0682	0.0701	0.0562	0.0573

user-item interaction graph. Clearly, the performance of our CaDSI with layer number $L = 2$ is better than that with $L = 1$, since the second-order connectivity can capture significant collaborative signals of users and items.

- When stacking more than 2 layers, the influence of multi-hop neighbors is small and the recommendation performance is degraded. This is reasonable since too informative signals of user-item interactions might introduce additional noises to the representation learning. This again emphasizes the importance of controlling the bias brought by context information.

4.4.4 Top- K Recommendation Performance

Based on the evaluation on Recall@ K and NDCG@ K , Fig. 7 shows that CaDSI achieves the stable performance on Top- K recommendation when K (i.e., the length of the ranking list) varies from 10 to 80. This indicates that our CaDSI performs stably on Top- K recommendation task and can recommend more relevant items within Top- K positions when the ranking list length increases.

4.5 Case Study and Visualization(RQ4)

We conduct experiments to get deep insights into the disentangled representations w.r.t. the disentanglement of the semantics of user intents, the representability and interpretability of the learned disentangled embedding. The case studies towards the disentanglement of the semantics of user intents are shown in Appendix A, the visualization results of the learned disentangled embeddings are shown in Appendix B.

5 RELATED WORK

In this section, we will introduce previous works related to ours from the following three aspects, including HIN enhanced representation, disentangled representation and causal inference for recommendation.

5.1 HIN Enhanced Representation

As a newly emerging direction, heterogeneous information network [14] is proved to be effective in modeling complex objects and providing rich semantics information to recommender systems [14], [33]. Many HIN-based recommendation methods achieve state-of-the-art performance [14]. For example, HeteMF [34] utilizes meta path-based similarities as regularization terms in the MF model. HeteRec [35]

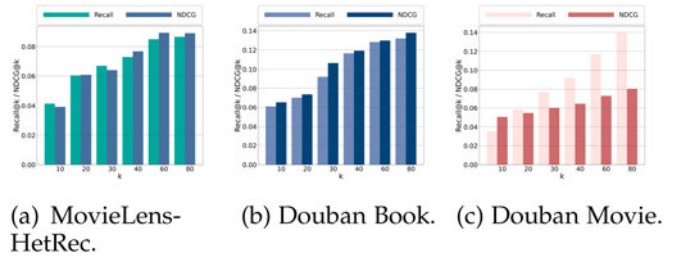


Fig. 7. Performance of CaDSI in terms of Recall@ K and NDCG@ K under different K .

learns meta path-based latent features based on different types of entity relationships and proposes an enhanced personalized recommendation framework. SemRec [36] proposes a weighted HIN and designs a meta path-based CF model to flexibly integrate heterogeneous information for a personalized recommendation. The effectiveness of HIN has been proved by a vast amount of HIN-based recommendation methods [14]; thus, in our work, we value HIN in providing rich semantic information of user and item types. Despite the effectiveness, neither the enhanced graph-based nor HIN-based representations can disentangle users' intents by just presuming uniform entangled embeddings behind behaviors. This can result in the poor interpretability of the developed recommendation methods. Thus, disentangle representation learning, which aims to learn factorized representations that separate and uncover latent explanatory factors behind the data [37], has recently received much attention in recommendation systems.

5.2 Disentangled Representation

Previous studies have demonstrated that disentangled representations are more robust, i.e., confounding biases are less likely to be preserved by uncovering latent factors. Ma *et al.* [8] propose to differentiate latent factors of learned user/item embeddings into macro and micro ones, thus the developed recommendation methods are less likely to preserve the confounding of the factors mistakenly. Moreover, the disentangle representation can provide rich semantics of users' preference, involving items' aspect information [8], [38] to users' behavior type information [39]. Thus, several works are proposed using disentangle representation learning to improve recommendation. For instance, DisHAN [40] learns disentangled aspect-aware user/item representations based on different meta path types in a HIN. Unfortunately, these aspect-aware disentangled embeddings only captured users' general taste on item aspects, however, failed to combine the specified item aspects with the real user intents. In the parallel, several works are conducted on modeling disentangled representations of users' intents, such as Macrid-VAE [8], DICE [11] and DGCf [31]. The drawbacks are also distinct: they failed to combine the learned user intents with real-world item aspects, i.e., the user intents are predefined manually, such as "passing the time". To sum up, the current studies on disentangle representation learning-enhanced recommendation either target at learning items' aspect-level representation [30], [40] or users' intents-level representation [8], [11], [31]. Moreover, aforementioned approaches ignore bias stemming from semantics information. To our knowledge, our approach is the first attempt to

achieve interpretable and unbiased recommendations with disentangled embeddings for user intent.

5.3 Causal Methods for Debiasing

To the best of our knowledge, existing causal methods for recommendations aim to mitigate the effects of different biases rather than improving interpretability as in our work. Most existing works claim that the observational rating data suffers from *selection bias* [9], [41], *exposure bias* [6], [7], [42], [43] or *popularity bias* [3], [4], [10]. Dominant approaches adopt two main strategies to remove these biases, namely, the propensity-score [5], [6], [42] and causal embedding [7], [41], [43]. For instance, the method in [6] uses propensity score to re-weight the observational click data, with the aim of imitating the scenario that item is randomly exposed and alleviating the *exposure bias*. The work in [7] learns a uniform unbiased embeddings from partially observed user-item interactions via their decoupled model. More recently, the work in [41] resorts to balance learning with a Middle-point Distance Minimization (MPDM) strategy to learn causal embeddings that are free from *selection bias*. Facing user conformity issue in recommendation, [3] relates such issue with *popularity bias* and proposes to alleviate the popularity bias by learning disentangled embeddings of user interest. A few state-of-the-art works [4], [9], [10] inspect cause-effect of the bias generation and design a specific causal graph attributing the *exposure bias* to a confounder. For example, Li *et al.* [9] prove the social network to be a confounder that affects the user's rating and the exposure mechanism of items to the user.

6 CONCLUSION AND FUTURE WORK

In this paper, we have researched the confounding bias issue stemming from different aspects, and propose an unbiased and robust *Causal Disentanglement semantic-aware Intent Learning* (CaDSI) for recommendation. Our CaDSI can provide semantics to fine-grained representations for disentangling user intents, meanwhile easing the bias stemming from unevenly distributed item aspects. We evaluate our CaDSI on three real-world recommendation datasets, with extensive experiments and visualizations demonstrate the robustness and interpretability of our semantic-aware user intent representation. In future work, we will explore the effect of different auxiliary information on the recommendation system using the intervention analysis in causal inference.

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Xiangmeng Wang and Qian Li have contributed equally to this work.

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Xiangmeng Wang received the MSc degree in computer application technology from Shanghai University. She has been working toward the PhD degree with the School of Computer Science, Faculty of Engineering and Information Technology, University of Technology Sydney (UTS). Her research interests include explainable artificial intelligence, data analysis, and causal machine learning. Her papers have been published in the top-tier conferences and journals in the field of machine learning.



Qian Li received the PhD degree in computer science from the Chinese Academy of Science. She is currently a lecturer with the School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University, Perth, Australia. She has been a postdoctoral research fellow with the School of Computer Science, Faculty of Engineering and Information Technology, University of Technology Sydney (UTS). Her research interests include optimization algorithms, topological data analysis, and causal machine learning. Her

papers have been published in the top-tier conferences and journals in the field of machine learning and computer vision.



Dianer Yu received the BSc degree of IT from the University of Technology Sydney. He has been working toward the postgraduate IT degree with the School of Computer Science, Faculty of Engineering and Information Technology, University of Technology Sydney (UTS). His research interests include data mining, causal model for recommendation and explainable machine learning. He has been awarded as postgraduate dean's List during the Postgraduate period.



Peng Cui (Senior Member, IEEE) received the PhD degree in computer science from Tsinghua University in 2010. He is currently an associate professor with Tsinghua University. He has vast research interests in data mining, multimedia processing, and social network analysis. Until now, he has published more than 20 papers in conferences such as SIGIR, AAAI, ICDM, etc. and journals such as *IEEE Transactions on Multimedia*, *IEEE Transactions on Image Processing*, *Data Mining and Knowledge Discovery*, etc. Now his research is sponsored by National Science Foundation of China, Samsung, Tencent, etc. He also serves as guest editor, co-chair, PC member, and reviewer of several high-level international conferences, workshops, and journals.



Zhichao Wang received the PhD degree from the Department of Automation, Tsinghua University. He was a research fellow with the University of New South Wales. His research interests include optimization for machine learning and stochastic modeling.



Guandong Xu (Member, IEEE) received the MSc and BSc degrees in computer science and Engineering, and the PhD degree in computer science. He is a professor with the School of Computer Science and Advanced Analytics Institute, University of Technology Sydney. He currently heads with the Data Science and Machine Intelligence Lab, which consists of 15+ members of academics, research fellows and HDR students. From Nov 2019, he directs the newly established Smart Future Research Centre,

which is an across-disciplines industry engagement and innovation platform for AI and Data Science Application towards smart wealth management and investment, energy, food, water, living, and city.

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